

# Heewon Lee

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## Education

### 2021 — Current: B.S. in Computer Science

KAIST, Korea | Double major: Physics | Minor: Materials Science and Engineering

- **GPA:** 4.10/4.30
- **Relevant Coursework:** Quantum Simulators, Phase Transitions, Thermal Physics, Quantum Mechanics II, Matrix Groups
- **Honors:** Dean's List (Spring 2021, Spring 2023, Fall 2023), Honor Program (Spring 2023 – Current)
- **Anticipated Graduation:** August 2027

### May 2024 — November 2025: Republic of Korea Army

Signal Intelligence & Electronic Warfare Operator

- Honorably discharged as Sergeant
- **Concurrent research** via remote guidance from Prof. Sukeyoung Ryu (June 2025 – February 2026)

## Research Experience

### Programming Systems Laboratory, KAIST

Undergraduate Intern, Advisor: Prof. Kihong Heo | June 2023 – April 2024

- Co-developed **UnitCon**, a static-analysis-driven **targeted unit test synthesizer** for Java programs.
- Focused on **test modification**, **bug discovery** in open-source projects, and **integrating static analysis reports** into the enumerative synthesizer pipeline.
- Co-authored research manuscript, culminating in a **Distinguished Paper Award at FSE 2025**.

### Independent Research

Advisor: Prof. Sukeyoung Ryu | June 2025 – February 2026 (During Military Service)

- Formalized **asymptotic analysis** as a pedagogical **abstract interpretation**.
- Proved soundness via **Galois connection** and developed **interactive tools** for classroom settings.
- Peer-reviewed and selected to present at the **ACM SIGPLAN POPL 2026 Student Research Competition**.

## Publications & Presentations

**UnitCon: Synthesizing Targeted Unit Tests for Java Runtime Exceptions.** Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE), 2025.

S. Jang, Y. Ryou, **H. Lee**, K. Heo [paper] [DOI]

🏆 **ACM SIGSOFT Distinguished Paper Award**

**Asymptotic Analysis as an Abstract Interpretation.** Poster presentation, ACM SIGPLAN POPL 2026 Student Research Competition (SRC).

**H. Lee**, S. Ryu [poster]

**Asymptotic Analysis: The Undergraduate's Abstract Interpretation.** Poster presentation, Special Interest Group on Programming Languages (SIGPL) Winter School 2026.

**H. Lee**, S. Ryu [poster in Korean]

## Skills

- **Research Interests:** Programming languages, formal semantics, quantum computation and information, quantum simulation
- **Languages:** Python, OCaml, Java, Lean 4
- **Quantum Software:** Qiskit, OpenQASM